

PROJECT JANUS MINI-16: 3D ELECTRO-PHOTONIC PEX EXTRACTION SIGN-OFF OVERLAY

Pre-Layout Nominal Budget (100 GHz) Physical Co-Cu Hybrid Pad (5.5 fF) + TDV Via (1.8 pF) + 35 μ m CPW Line vs Nominal Budget (100 GHz)

$C_{load} = 28.5$ fF $f_{3dB} = 175.0$ GHz Opening = 87.3%

Post-Layout 3D PEX Extracted (100 GHz)

$C_{load} = 31.99$ fF $f_{3dB} = 166.1$ GHz $\Delta Loss = -3.07\%$

